CSE 309 Automata
Spring 2026
Midterm Exam
10 March 2026
Time Limit: 120 Minutes

Name: SAMPLE SOLUTION

Student ID: ____________________

This exam contains 6 pages (including this cover page) and 5 problems.

- **DON'T PANIC.** Tackle everything on the test. Don't skip questions because they look "scary": once you figure out what they're asking, you'll probably realize they're a lot easier than they seemed!
- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by logical reasoning, explanation, or algebraic work will receive no credit if incorrect; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- If you need more space, use the back of the pages; clearly indicate when you have done this.

Problem	Points	Score
1	10	
2	5	
3	10	
4	10	
5	10	
Total:	45	

Do not write in the table to the right.

1. (10 points) If A is any language, let B be the set of all first halves of strings in A, so that

$$B = \{x \mid \text{for some } y,\ |x| = |y| \text{ and } xy \in A\}.$$

Show that if A is regular, then so is B.

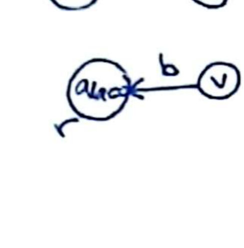
Assume NFA M recognizes A with single accept state.

$M = (Q, \Sigma, \delta, q_0, q_{acc})$

$N = (Q', \Sigma, \delta', q_0', F')$ for B is as follows

- $Q' = Q \times Q$
- For $q, r \in Q$, define
  $\delta'((q,r), a) = \{(u,v) \mid u \in \delta(q,a)$ & $r \in \delta(v,b)$ for some $b \in \Sigma\}$
- $q_0' = (q_0, q_{acc})$
- $F' = \{(q,q) \mid q \in Q\}$

e.g. Initially



2. (5 points) If A is any language, let C be the set of all strings in A with their middle thirds removed, so that

$$C = \{xz \mid \text{for some } y,\ |x| = |y| = |z| \text{ and } xyz \in A\}.$$

Show that if A is regular, then C is not necessarily regular. *Hint: Use closure properties.*

$A = \{0^* \# 1^*\}$

$C \cap \{0^* 1^*\} = \{0^n 1^n \mid n \geq 0\}$

4. (10 points) Give a context-free grammar for the complement of the language $\{a^n b^n \mid n \geq 0\}$.

$S \rightarrow XbXaX \mid T \mid U$

$T \rightarrow aTb \mid Tb \mid b$

$U \rightarrow aUb \mid aU \mid a$

$X \rightarrow aX \mid bX \mid \lambda$

3. (10 points) Let $\Sigma = \{a\}$. Show via pumping lemma for regular languages that the language

$$L = \{a^{n^2} \mid n \geq 0\}$$

is not regular.

Let n be pumping length.

Choose $w = a^{n^2}$

$|w| = n^2 > n$

$w_i = xy^iz = a^{n^2+(i-1)k}$, $y = a^k$, $1 \leq k \leq n$

$i = 2 \rightarrow n^2 + k$ cannot be a perfect square.

since $(n+1)^2 = n^2 + 2n + 1$

$n^2 + 2n + 1 > n^2 + k$

5. (10 points) *Infinite pancakes all the way!* Imagine that our friend Sheikh Chilly is capable of solving the halting problem $HALT$ via a black box. Given any instance $\langle M, w \rangle$ to the halting problem, we can query the box if $\langle M, w \rangle \in HALT$, and it will instantaneously give us the right answer.

He is willing to provide us the box for free. Show that there still exist problems that are unsolvable even after using the box as an oracle by explicitly constructing such a problem. Use your construction to argue that there exists an infinite hierarchy of unsolvable problems.

Define Super HALT for TMs that have access to oracle. Similar contradiction can be constructed for Super HALT.